

## CERBERUS, Australia

WMO: 948980

Lat: 38.3647S

Lon: 145.1786E

Elev: 14

StdP: 101.16

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 7             | 2.1        | 3.5 | 0.6                           | 4.0 | 7.4  | 1.7 | 4.3 | 8.1  | 11.1                  | 12.1 | 10.1 | 11.7 | 0.8                      | 0    | 0.514 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 9.3                    | 33.0            | 18.9 | 30.1 | 18.6 | 27.3 | 18.1 | 21.0                | 27.5 | 20.1 | 26.2 | 19.3 | 25.0 | 5.4                  | 340  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.2                            | 14.0 | 22.6 | 18.2 | 13.2 | 22.0 | 17.3 | 12.4 | 21.4 | 61.0          | 27.2 | 57.8 | 26.3 | 54.9 | 24.9 | 25.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 9.9           | 8.8  | 7.9 | DB                         | -0.3 | 38.6               | 1.1 | 2.7                                                | -1.1 | 40.6       | -1.7 | 42.1       | -2.4 | 43.6       | -3.2 | 45.6 | (4) |
| (5)               |      |     | WB                         | -0.7 | 22.9               | 1.1 | 0.9                                                | -1.5 | 23.6       | -2.1 | 24.1       | -2.8 | 24.7       | -3.6 | 25.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 14.1        | 18.8  | 19.2  | 17.5  | 14.6  | 12.3  | 10.3  | 9.8   | 10.4  | 11.7  | 13.3  | 15.2  | 16.7  | (6)  |  |
|                                                                        | DBStd       | 4.30        | 3.72  | 3.39  | 3.30  | 2.77  | 2.26  | 2.12  | 1.95  | 2.10  | 2.45  | 2.91  | 3.25  | 3.56  | (7)  |  |
|                                                                        | HDD10.0     | 82          | 0     | 0     | 0     | 1     | 4     | 22    | 26    | 19    | 8     | 2     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 1716        | 38    | 27    | 57    | 116   | 186   | 241   | 263   | 246   | 200   | 159   | 107   | 77    | (9)  |  |
|                                                                        | CDD10.0     | 1591        | 273   | 256   | 233   | 140   | 77    | 31    | 22    | 32    | 58    | 105   | 156   | 209   | (10) |  |
|                                                                        | CDD18.3     | 183         | 52    | 50    | 31    | 6     | 0     | 0     | 0     | 0     | 1     | 4     | 13    | 28    | (11) |  |
|                                                                        | CDH23.3     | 1766        | 493   | 414   | 269   | 49    | 1     | 0     | 0     | 0     | 6     | 69    | 152   | 312   | (12) |  |
| (13)                                                                   | CDH26.7     | 724         | 237   | 169   | 101   | 9     | 0     | 0     | 0     | 0     | 0     | 17    | 49    | 143   | (13) |  |
| (14)                                                                   | Wind        | WSAvg       | 3.6   | 3.9   | 3.8   | 3.5   | 3.1   | 3.2   | 3.2   | 3.7   | 3.8   | 3.8   | 3.7   | 3.9   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 776         | 43    | 47    | 48    | 70    | 72    | 75    | 77    | 79    | 75    | 69    | 68    | 51    | (15) |  |
|                                                                        | PrecMax     | 990         | 125   | 152   | 80    | 196   | 160   | 148   | 151   | 138   | 144   | 118   | 137   | 119   | (16) |  |
|                                                                        | PrecMin     | 521         | 4     | 8     | 18    | 21    | 29    | 16    | 37    | 20    | 29    | 15    | 25    | 3     | (17) |  |
|                                                                        | PrecStd     | 131         | 30    | 32    | 17    | 41    | 33    | 32    | 28    | 31    | 32    | 30    | 29    | 27    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 37.9  | 35.3  | 33.6  | 27.9  | 21.9  | 17.5  | 16.9  | 20.1  | 24.0  | 29.0  | 31.3  | 35.3  | (19) |  |
|                                                                        |             | MCWB        | 20.3  | 19.6  | 18.5  | 16.1  | 14.1  | 12.7  | 11.3  | 12.8  | 14.5  | 16.2  | 18.2  | 18.9  | (20) |  |
|                                                                        | 2%          | DB          | 32.7  | 31.6  | 29.4  | 24.5  | 19.5  | 15.9  | 15.2  | 17.5  | 20.7  | 25.0  | 27.7  | 31.0  | (21) |  |
|                                                                        |             | MCWB        | 19.3  | 19.3  | 18.0  | 15.3  | 13.6  | 12.3  | 10.9  | 11.6  | 13.2  | 15.0  | 17.5  | 18.0  | (22) |  |
|                                                                        | 5%          | DB          | 28.6  | 28.4  | 26.2  | 22.0  | 18.0  | 14.7  | 14.1  | 15.8  | 18.7  | 22.1  | 24.3  | 26.5  | (23) |  |
|                                                                        |             | MCWB        | 19.2  | 19.2  | 17.7  | 15.3  | 13.2  | 11.7  | 10.4  | 11.0  | 12.7  | 14.7  | 16.9  | 17.8  | (24) |  |
|                                                                        | 10%         | DB          | 25.1  | 25.3  | 23.3  | 19.6  | 16.4  | 13.8  | 13.2  | 14.3  | 16.7  | 19.2  | 21.2  | 23.0  | (25) |  |
|                                                                        |             | MCWB        | 18.9  | 18.8  | 17.3  | 14.8  | 12.7  | 11.1  | 10.1  | 10.4  | 11.9  | 13.7  | 16.1  | 17.5  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 22.5  | 22.4  | 20.9  | 17.8  | 15.7  | 14.3  | 12.7  | 13.7  | 15.9  | 17.7  | 20.4  | 20.7  | (27) |  |
|                                                                        |             | MCDB        | 28.7  | 28.9  | 27.8  | 23.4  | 18.7  | 15.8  | 15.0  | 17.5  | 20.5  | 23.8  | 26.9  | 28.8  | (28) |  |
|                                                                        | 2%          | WB          | 21.1  | 21.0  | 19.5  | 16.8  | 14.7  | 12.9  | 11.7  | 12.4  | 14.3  | 16.3  | 18.6  | 19.5  | (29) |  |
|                                                                        |             | MCDB        | 27.4  | 27.0  | 25.0  | 21.0  | 17.5  | 14.9  | 13.8  | 16.0  | 19.1  | 22.4  | 24.9  | 26.3  | (30) |  |
|                                                                        | 5%          | WB          | 20.1  | 20.1  | 18.6  | 16.0  | 13.9  | 12.1  | 11.0  | 11.5  | 13.3  | 15.2  | 17.5  | 18.5  | (31) |  |
|                                                                        |             | MCDB        | 26.6  | 25.3  | 23.7  | 20.0  | 16.9  | 14.2  | 13.2  | 14.6  | 17.4  | 20.5  | 22.7  | 24.9  | (32) |  |
|                                                                        | 10%         | WB          | 19.1  | 19.2  | 17.7  | 15.2  | 13.2  | 11.4  | 10.5  | 10.8  | 12.3  | 14.1  | 16.3  | 17.4  | (33) |  |
|                                                                        |             | MCDB        | 24.8  | 24.2  | 22.6  | 18.9  | 15.9  | 13.5  | 12.6  | 13.6  | 15.9  | 18.5  | 20.6  | 22.9  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.5   | 9.3   | 9.3   | 8.9   | 7.4   | 6.6   | 6.4   | 7.1   | 8.2   | 9.0   | 8.9   | 9.6   | (35) |  |
|                                                                        |             | MCDBR       | 17.5  | 15.8  | 15.3  | 13.2  | 10.1  | 7.7   | 7.3   | 9.9   | 12.2  | 15.1  | 15.7  | 17.5  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 5.8   | 5.7   | 5.7   | 6.0   | 5.6   | 5.1   | 4.5   | 5.4   | 6.0   | 6.8   | 6.9   | 6.8   | (37) |  |
|                                                                        |             | MCWBR       | 13.5  | 11.9  | 12.3  | 10.9  | 8.7   | 7.0   | 6.6   | 8.5   | 10.4  | 13.0  | 13.0  | 14.0  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.358 | 0.358 | 0.336 | 0.333 | 0.319 | 0.308 | 0.310 | 0.322 | 0.343 | 0.344 | 0.350 | 0.346 | (40) |  |
|                                                                        |             | taud        | 2.505 | 2.515 | 2.579 | 2.561 | 2.578 | 2.598 | 2.583 | 2.543 | 2.480 | 2.495 | 2.505 | 2.532 | (41) |  |
|                                                                        | Ebn at Noon |             | 973   | 949   | 931   | 870   | 820   | 802   | 823   | 868   | 907   | 953   | 976   | 990   | (42) |  |
|                                                                        |             | Edh at Noon | 112   | 106   | 92    | 82    | 70    | 63    | 68    | 82    | 100   | 107   | 111   | 110   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 7.04  | 6.16  | 4.85  | 3.37  | 2.23  | 1.76  | 1.98  | 2.80  | 4.01  | 5.38  | 6.31  | 7.04  | (44) |  |
|                                                                        | RadStd      |             | 0.42  | 0.30  | 0.27  | 0.18  | 0.12  | 0.13  | 0.09  | 0.15  | 0.25  | 0.38  | 0.34  | 0.39  | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) | Station Only            | +0.63 | N/A     | N/A    | N/A     | N/A   | N/A   | -28         | -199    | +227    | +50     | (46) |  |
| (47) | Regional (33 neighbors) | +0.50 | N/A     | N/A    | +0.79   | N/A   | N/A   | -17         | -132    | +174    | +57     | (47) |  |

Nomenclature: See separate page